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Subject: Programming with JAVA

Sem: III

AY: 2026-27

Assignment No: 18

TITLE: Write a Java GUI program to demonstrate button click event handling.

Problem Statement

Write a Java GUI program to demonstrate button click event handling.

Learning Objectives

To implement button, click events using Java event handling mechanisms.

Theory

Event handling allows Java programs to respond to user actions such as button clicks, mouse movements, and keyboard input. Swing uses ActionListener to process button click events. Event-driven programming separates user interface design from event-processing logic, making applications interactive and user-friendly.

Exercise

1. Create a GUI calculator where buttons perform addition and subtraction.

```
import javax.swing.*;
```

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class Calculator implements ActionListener {
```

```
    JFrame frame;
```

```
    JTextField num1, num2, result;
```

```
    JButton add, subtract;
```

```
Calculator() {  
  
    frame = new JFrame("GUI Calculator");  
  
  
    JLabel label1 = new JLabel("First Number:");  
    JLabel label2 = new JLabel("Second Number:");  
    JLabel label3 = new JLabel("Result:");  
  
  
    num1 = new JTextField();  
    num2 = new JTextField();  
    result = new JTextField();  
    result.setEditable(false);  
  
  
    add = new JButton("Addition");  
    subtract = new JButton("Subtraction");  
  
  
    label1.setBounds(40, 40, 120, 30);  
    num1.setBounds(160, 40, 150, 30);  
  
    label2.setBounds(40, 80, 120, 30);  
    num2.setBounds(160, 80, 150, 30);  
  
    label3.setBounds(40, 120, 120, 30);  
    result.setBounds(160, 120, 150, 30);  
  
  
    add.setBounds(70, 170, 110, 35);  
    subtract.setBounds(200, 170, 110, 35);  
  
  
    frame.add(label1);  
    frame.add(num1);  
    frame.add(label2);
```

```

frame.add(num2);

frame.add(label3);

frame.add(result);

frame.add(add);

frame.add(subtract);


add.addActionListener(this);

subtract.addActionListener(this);


frame.setSize(380, 270);

frame.setLayout(null);

frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

frame.setVisible(true);
}


public void actionPerformed(ActionEvent e) {

    double a = Double.parseDouble(num1.getText());

    double b = Double.parseDouble(num2.getText());


    if (e.getSource() == add) {

        result.setText(String.valueOf(a + b));

    } else if (e.getSource() == subtract) {

        result.setText(String.valueOf(a - b));

    }

}


public static void main(String[] args) {

    new Calculator();

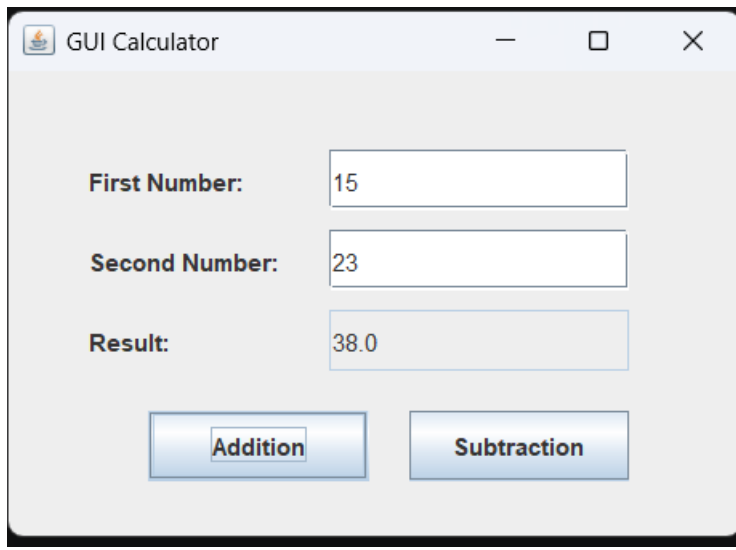
}

}

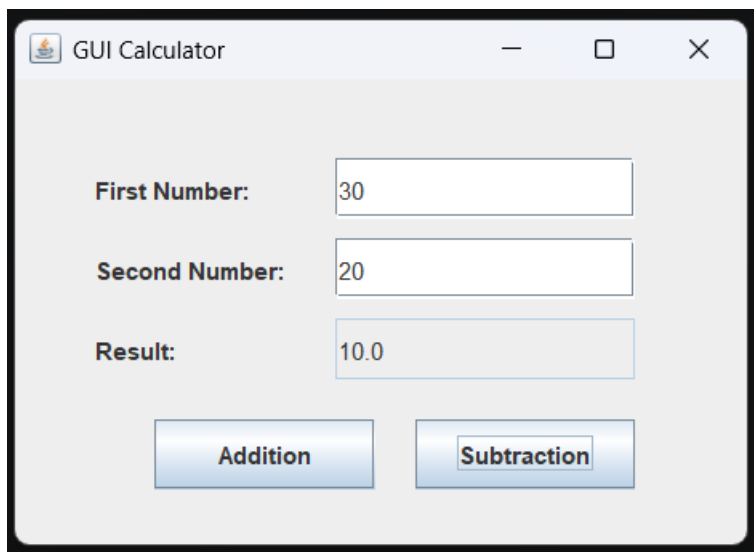
```

OUTPUT:

Performing Addition



Performing Subtraction



2. Create a Bank Balance Calculator GUI application using Java Swing where the user enters the initial balance and transaction amount. Use buttons to perform addition (deposit) and subtraction (withdrawal) and display the updated balance.

```
import javax.swing.*;
```

```
import java.awt.event.*;
```

```
public class BankBalanceCalculator implements ActionListener {
```

```
    JFrame frame;
```

```
    JTextField balanceField, amountField, resultField;
```

```
JButton depositButton, withdrawButton;
```

```
BankBalanceCalculator() {
```

```
    frame = new JFrame("Bank Balance Calculator");
```

```
    JLabel balanceLabel = new JLabel("Initial Balance:");
```

```
    JLabel amountLabel = new JLabel("Transaction Amount:");
```

```
    JLabel resultLabel = new JLabel("Updated Balance:");
```

```
    balanceField = new JTextField();
```

```
    amountField = new JTextField();
```

```
    resultField = new JTextField();
```

```
    resultField.setEditable(false);
```

```
    depositButton = new JButton("Deposit");
```

```
    withdrawButton = new JButton("Withdraw");
```

```
    balanceLabel.setBounds(40, 40, 130, 30);
```

```
    balanceField.setBounds(180, 40, 180, 30);
```

```
    amountLabel.setBounds(40, 80, 130, 30);
```

```
    amountField.setBounds(180, 80, 180, 30);
```

```
    resultLabel.setBounds(40, 120, 130, 30);
```

```
    resultField.setBounds(180, 120, 180, 30);
```

```
    depositButton.setBounds(80, 170, 110, 35);
```

```
    withdrawButton.setBounds(220, 170, 110, 35);
```

```
    frame.add(balanceLabel);
```

```

frame.add(balanceField);
frame.add(amountLabel);
frame.add(amountField);
frame.add(resultLabel);
frame.add(resultField);
frame.add(depositButton);
frame.add(withdrawButton);

depositButton.addActionListener(this);
withdrawButton.addActionListener(this);

frame.setSize(420, 270);
frame.setLayout(null);
frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
frame.setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    double balance = Double.parseDouble(balanceField.getText());
    double amount = Double.parseDouble(amountField.getText());

    if (e.getSource() == depositButton) {
        balance = balance + amount;
    } else if (e.getSource() == withdrawButton) {
        balance = balance - amount;
    }

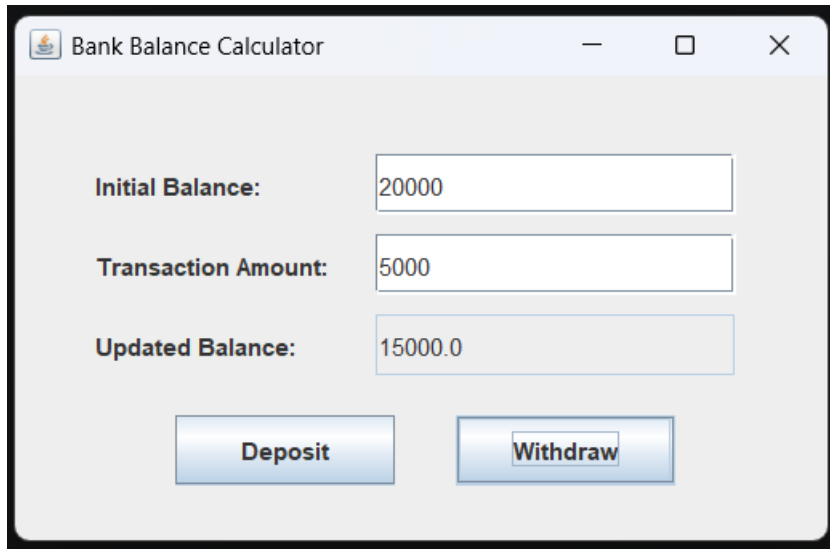
    resultField.setText(String.valueOf(balance));
}

```

```
public static void main(String[] args) {  
    new BankBalanceCalculator();  
}  
}
```

OUTPUT:

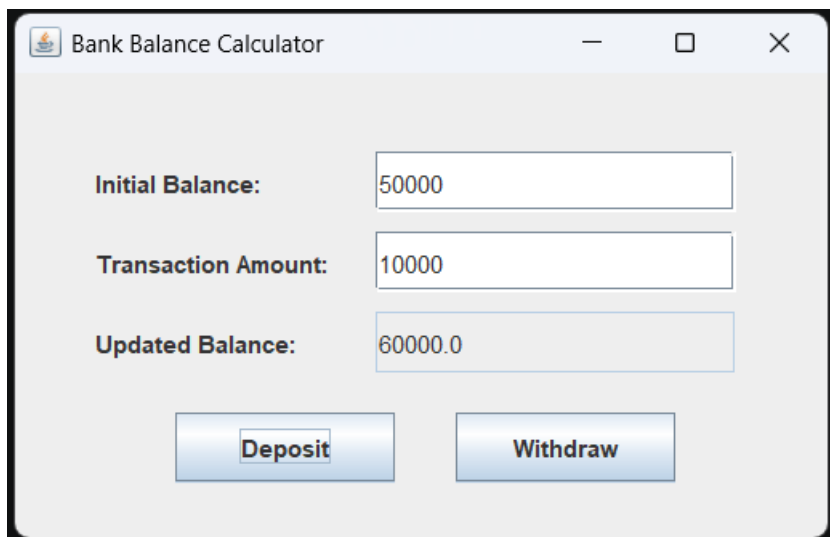
Performing withdrawal



The screenshot shows a Java Swing window titled "Bank Balance Calculator". It contains three text input fields and two buttons. The "Initial Balance" field contains the value "20000". The "Transaction Amount" field contains the value "5000". The "Updated Balance" field, which is disabled, displays the result "15000.0". Below the fields are two buttons: "Deposit" and "Withdraw". The "Withdraw" button is highlighted, indicating it was the last active component.

Field	Value
Initial Balance:	20000
Transaction Amount:	5000
Updated Balance:	15000.0

Performing Deposit



The screenshot shows the same "Bank Balance Calculator" window. In this state, the "Initial Balance" field contains "50000", the "Transaction Amount" field contains "10000", and the disabled "Updated Balance" field displays "60000.0". The "Deposit" button is highlighted, indicating it was the last active component.

Field	Value
Initial Balance:	50000
Transaction Amount:	10000
Updated Balance:	60000.0

Conclusion

We learned how to handle button click events using Java Swing and ActionListener. The calculator and bank balance applications successfully perform operations and display the updated results based on user actions.